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Problem

BEYOND HUMAN COT

- Reasoning gains often depend on expensive human-written chains of thought and supervised fine-tuning.
- Imitating demonstrations can transmit human bias and cap exploration of alternative reasoning paths.
- Can a pretrained model discover stronger reasoning from outcome feedback alone?

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R1-Zero

PURE RL SELF-EVOLUTION

- Start from DeepSeek-V3 Base, skip SFT, and optimize with Group Relative Policy Optimization (GRPO).
- Rule-based accuracy and format rewards constrain the answer, not the internal reasoning strategy.
- Reasoning spontaneously lengthens and develops verification, reflection, alternative search and 'aha' rethinking.

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DeepSeek-R1

MULTISTAGE ALIGNMENT

- Cold-start reasoning data improves readability and language consistency before the first RL stage.
- Rejection sampling plus SFT mixes reasoning, writing and software-engineering data.
- A second RL stage combines verifiable rewards with helpfulness and safety preference rewards.

DEEPSEEK-R1

DeepSeek-R1

Incentivizing Reasoning through Reinforcement Learning | Nature 2025

Large-scale reinforcement learning with reliable, verifiable rewards can unlock self-evolving reasoning—reflection, verification and strategy switching—without human-labelled reasoning traces.

GRPOreasoning RLdistillation

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Optimization

RELATIVE ADVANTAGE

- $A_i = (r_i - \text{mean}(r_1 \dots r_G)) / \text{std}(r_1 \dots r_G)$
- GRPO compares a group of sampled answers and avoids the separate value model used by PPO.
- Distillation transfers emergent long-CoT behavior to smaller dense models better than small-model RL alone.

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Evidence

REASONING EMERGES

- R1-Zero AIME 2024 pass@1 rises from 15.6% to 77.9%; self-consistency reaches 86.7%.
- Final R1 reaches AIME 79.8%, Codeforces rating 2,029, LiveCodeBench 65.9% and MATH-500 97.3%.
- General alignment also improves: IF-Eval 83.3, AlpacaEval 87.6 and Arena-Hard 92.3.

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Limits & meaning

RELIABLE VERIFIERS MATTER

- Pure RL works best where correctness can be checked; weak learned rewards invite reward hacking.
- R1 still shows overthinking, prompt sensitivity, language mixing, limited tool use and moderate intrinsic safety.
- The central recipe is hard problems + a reliable verifier + enough RL compute, not massive reasoning annotation.